

Final Year Project Proposal Form

Please complete this form and have your project proposal approved by your proposed supervisor. In addition, you are required to read and sign the declarations at the bottom of the form. Submit your completed form via Brightspace.
Digital supervisor signatures are acceptable.

Name	Julia Hooper
Student ID	24430706
Supervisor Name	Colin Venters
Project Title *(initial title or working title)*	Screening Training Data for Representational Bias Before LLM Training: A CI-Integrated, Self-Hostable Tool

Description of Own Proposed Project
It should contain a detailed description of the project and should include sections on 'Background Research to Date', 'Key Aims', 'Special Resources Required' and 'Significant Milestones'. This description should be at least 500 words long.

This project proposes the design, implementation and evaluation of an open-source, self-hostable tool that scans training data, comprising scraped web content and structured datasets, for bias and harmful stereotyping before it enters an LLM training pipeline. The tool would run as a CI/CD step, scoring each new source and accepting or rejecting it against developer-configured thresholds. Medical text is the principal case study used to validate the tool. The same scanner could also help dataset and website publishers to self-assess their content before release.

Background Research to Date

Existing curation toolkits, including NVIDIA NeMo Curator, Dolma and datatrove, support deduplication and toxicity filtering for large, one-off pretraining runs. They are not designed as lightweight, per-source gates for small teams, and toxicity filters are unlikely to detect subtler representational bias, such as stereotyped or dismissive descriptions of particular groups. Commercial governance platforms, such as Credo AI and Holistic AI, offer bias auditing and EU AI Act oriented workflows, but they are enterprise products rather than open, self-hostable detectors. Bias research has largely examined model outputs, for example GenderBench (Pikuliak 2025) and studies reporting gender-dependent urgency in medical LLM responses. This project

instead addresses the training data upstream. Benchmarks such as StereoSet and CrowS-Pairs will be reviewed for validation, and the accuracy cost of running a local model rather than a commercial API will be measured.

Key Aims

Aim: To design, implement and evaluate an open-source, self-hostable tool that screens training data for representational bias and stereotyping, as a CI/CD step before the data is admitted to an LLM training pipeline.

Objective 1: To review existing curation tools, governance platforms and bias benchmarks in order to define a measurable notion of representational bias and to establish the gap the tool addresses.

Objective 2: To develop a detector for subtle representational bias by fine-tuning a small open-weight model that runs locally, with an optional commercial API backend and an ensemble mode that flags disagreements for human review.

Objective 3: To implement a CI/CD-integrated gate with developer-configurable score thresholds that accept or reject each source, and a report that non-technical staff can act on.

Objective 4: To validate the tool on medical text, using existing labelled benchmarks and a small labelled sample built around documented medical biases, and to benchmark the backends on accuracy, cost and privacy.

Objective 5: To evaluate report usability, through readability measures, a heuristic evaluation and a cognitive walkthrough, and to analyse and report the findings.

Special Resources Required

- A laptop and GPU access for fine-tuning, using free cloud notebooks or university compute
- Public datasets with verified licences
- Free or low-cost access to one or two commercial APIs
- GitHub Actions for CI integration
- Supervisory guidance on medical text and bias labelling.

Significant Milestones

By 25/10/2026 (interim report): scope and validation plan agreed with the supervisor, ethics position confirmed, literature review drafted and baseline results obtained on benchmark data.

November to December 2026: development of the scanning pipeline and thresholds, fine-tuning of the local model and a first working CI integration.

January to February 2027: addition of the commercial backend and ensemble mode, construction of the medical labelled sample and execution of the main benchmark.

March 2027: evaluation of report usability and analysis of results.

April 2027: IEEE-format write-up, 2 to 3 minute video summary and final submission

Reference

Pikuliak, M. (2025) 'GenderBench: evaluation suite for gender biases in LLMs', arXiv:2505.12054,

available: https://arxiv.org/abs/2505.12054 [accessed 07 Oct 2026].

Declarations

Sign-off by Proposed Supervisor

Supervisor Signature *(Proposed Supervisor)*	
Supervisor Name	Colin
Date	08/10/2026

Declaration 2: Ethical Approval

I declare that I have understood that undertaking a project involving human participants, including but not limited to interviews, user testing or questionnaires will require approval from the Faculty of Science and Engineering Ethics Committee. While it is the responsibility of the supervisor to lead and guide the application process, it is the student's responsibility to plan for sufficient time for this process. The lead-time on this is generally four to six weeks. Further details will be in a FYP lecture.	
Student Signature	Julia Hooper
Date	24/01/2006